Math 344, F18 **Name:** Answer
Exam 2

Instructions:

- There are totally 5 problems.
- You must show all of your work to receive a credit for a correct answer.
- No calculator is allowed in the quiz and exam.

Problem	Points	Score
1	24	
2	20	
3	22	
4	16	
5	18	
Total	100	

Good Luck !

Problem 1. (24 points) Let $\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 8 & 10 \\ 3 & 6 & 7 & 10 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 0 \\ -4 \\ 4 \end{bmatrix}$.

(a) Reduce $[\mathbf{A} \;\; \mathbf{b}]$ to $[\mathbf{R} \;\; \mathbf{d}]$ where $\mathbf{R}$ denotes the *reduced echelon form* of $\mathbf{A}$.

$$[A \vdots b] = \left[\begin{array}{cccc|c} 1 & 2 & 3 & 4 & 0 \\ 2 & 4 & 8 & 10 & -4 \\ 3 & 6 & 7 & 10 & 4 \end{array}\right] \to \left[\begin{array}{cccc|c} 1 & 2 & 3 & 4 & 0 \\ 0 & 0 & 2 & 2 & -4 \\ 0 & 0 & -2 & -2 & 4 \end{array}\right]$$

$$\to \left[\begin{array}{cccc|c} 1 & 2 & 3 & 4 & 0 \\ 0 & 0 & 2 & 2 & -4 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \to \left[\begin{array}{cccc|c} 1 & 2 & 3 & 4 & 0 \\ 0 & 0 & 1 & 1 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

$$\to \left[\begin{array}{cccc|c} 1 & 2 & 0 & 1 & 6 \\ 0 & 0 & 1 & 1 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] = [R \vdots d]$$

(b) Find a particular solution $\mathbf{x}_p$ of $\mathbf{Ax} = \mathbf{b}$ by setting the free variables to be zero.

$$Ax_p = b \Leftrightarrow Rx_p = d \Rightarrow \begin{cases} x_1 + 2x_2 + x_4 = 6 \\ x_3 + x_4 = -2 \end{cases}$$

Free variables: x_2, x_4 → setting $x_2 = x_4 = 0$

$$\Rightarrow \begin{cases} x_1 = 6 \\ x_2 = -2 \end{cases} \Rightarrow x_p = \begin{bmatrix} 6 \\ 0 \\ -2 \\ 0 \end{bmatrix}$$

(c) Find the complete solution to $\mathbf{Ax} = \mathbf{b}$.

Solve $Ax_n = 0 \Leftrightarrow Rx_n = 0$

$$\begin{cases} x_1 + 2x_2 + x_4 = 0 \\ x_3 + x_4 = 0 \end{cases}$$

Pivot variables: x_1, x_3

i) set $x_2 = 1$, $x_4 = 0$, we get $\begin{cases} x_1 = -2 \\ x_3 = 0 \end{cases}$

$\Rightarrow s_1 = (-2, 1, 0, 0)$

ii) set $x_2 = 0$, $x_4 = 1$, we get $\begin{cases} x_1 = -1 \\ x_3 = -1 \end{cases}$

$\Rightarrow s_2 = (-1, 0, -1, 1)$

Thus $x = x_p + x_n = \begin{bmatrix} 6 \\ 0 \\ -2 \\ 0 \end{bmatrix} + c_1 \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 0 \\ -1 \\ 1 \end{bmatrix}$

Problem 2. (20 points)

Let $\mathbf{A} = \begin{bmatrix} 2 & 4 & 0 & 5 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$.

(a) Find the dimension and a basis for the row space of $\mathbf{A}$, $C(\mathbf{A}^T)$.

$\dim(C(A^T)) = 2 \leftarrow$ Pivot variables : X_1, X_3

a basis: $\left\{ \begin{bmatrix} 2 \\ 4 \\ 0 \\ 5 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 2 \end{bmatrix} \right\}$

(b) Find the dimension and a basis for the column space of $\mathbf{A}$, $C(\mathbf{A})$.

$\dim(C(A)) = 2$

a basis: $\left\{ \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$

(c) Find the dimension and a basis for the nullspace of $\mathbf{A}$, $N(\mathbf{A})$.

Free variables : $X_2, X_4 \Rightarrow \dim(N(A)) = 2$

Solve $AX = 0$, i.e. $\begin{cases} 2X_1 + 4X_2 + 5X_4 = 0 \\ X_3 + 2X_4 = 0 \end{cases}$

i) Set $X_2 = 1, X_4 = 0$, we get $X_1 = -2, X_3 = 0$

$\Rightarrow S_1 = (-2, 1, 0, 0)$

ii) Set $X_2 = 0, X_4 = 1$, we get $X_1 = -\frac{5}{2}, X_3 = -2$

$\Rightarrow S_2 = (-\frac{5}{2}, 0, -2, 1)$

$\Rightarrow$ a basis: $\left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -5/2 \\ 0 \\ -2 \\ 1 \end{bmatrix} \right\}$

(d) Find the dimension and a basis for the left nullspace of $\mathbf{A}$, $N(\mathbf{A}^T)$.

$N(A^T) = 3 - \dim(C(A)) = 3 - 2 = 1$

Solve $A^T y = 0 \Rightarrow \begin{bmatrix} 2 & 0 & 0 \\ 4 & 0 & 0 \\ 1 & 0 & 0 \\ 5 & 2 & 0 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = 0$

$\Rightarrow \begin{cases} 2y_1 = 0 \\ 4y_2 = 0 \\ y_2 = 0 \\ 5y_1 + 2y_2 = 0 \end{cases} \Rightarrow \begin{matrix} y_1 = 0 \\ y_2 = 0 \end{matrix} \Rightarrow y = \begin{bmatrix} 0 \\ 0 \\ c \end{bmatrix} = c \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

$\Rightarrow$ a basis: $\left\{ \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$

Problem 3. (22 points)
(a) Find the straight line $C + Dt$ that comes closest to the values $\mathbf{b} = 0, 8, 8, 20$ at times $t = 0, 1, 3, 4$ by the least squares fitting.

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 3 \\ 1 & 4 \end{bmatrix}, \quad x = \begin{bmatrix} C \\ D \end{bmatrix}, \quad b = \begin{bmatrix} 0 \\ 8 \\ 8 \\ 10 \end{bmatrix}$$

$$A^TA = \begin{bmatrix} 4 & 8 \\ 8 & 26 \end{bmatrix}, \quad A^Tb = \begin{bmatrix} 36 \\ 112 \end{bmatrix}$$

Solve $A^TA\hat{x} = A^Tb$

$$\begin{bmatrix} 4 & 8 \\ 8 & 26 \end{bmatrix}\begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} 36 \\ 112 \end{bmatrix} \Rightarrow C = 1, \ D = 4$$

$\Rightarrow$ The least squares fitting line is

$$y = 1 + 4t$$

(a) Compute $\det(\mathbf{A})$ by reducing $\mathbf{A}$ to an upper triangular form $\mathbf{U}$ where

$$\mathbf{A} = \begin{bmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{bmatrix}.$$

$$A = \begin{bmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & -1 & 0 & 0 \\ 0 & 3/2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & -1 & 0 & 0 \\ 0 & 3/2 & -1 & 0 \\ 0 & 0 & \frac{4}{3} & -1 \\ 0 & 0 & -1 & 2 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 2 & -1 & 0 & 0 \\ 0 & 3/2 & -1 & 0 \\ 0 & 0 & 4/3 & -1 \\ 0 & 0 & 0 & 5/4 \end{bmatrix}$$

$$\Rightarrow \det A = 2 \cdot \left(\tfrac{3}{2}\right) \cdot \left(\tfrac{4}{3}\right) \cdot \left(\tfrac{5}{4}\right) = 5$$

Problem 4. (16 points)

Let $\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \\ 2 & 0 & 4 \end{bmatrix}$. Find an orthonormal basis $\mathbf{q}_1$, $\mathbf{q}_2$, $\mathbf{q}_3$ for the column space of $\mathbf{A}$ using the **Gram-Schmidt** process.

$$\tilde{q}_1 = a_1 = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \quad \|\tilde{q}_1\| = \sqrt{1^2+1^2+2^2} = \sqrt{6}$$

$$\Rightarrow q_1 = \frac{\tilde{q}_1}{\|\tilde{q}_1\|^2} = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} / \sqrt{6} = \begin{bmatrix} 1/\sqrt{6} \\ 1/\sqrt{6} \\ 2/\sqrt{6} \end{bmatrix}$$

$$\tilde{q}_2 = a_2 - (a_2^T q_1) q_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} - \left(\begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 1/\sqrt{6} \\ 1/\sqrt{6} \\ 2/\sqrt{6} \end{bmatrix} \right) \begin{bmatrix} 1/\sqrt{6} \\ 1/\sqrt{6} \\ 2/\sqrt{6} \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} - 0 \cdot \begin{bmatrix} 1/\sqrt{6} \\ 1/\sqrt{6} \\ 2/\sqrt{6} \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \quad \|\tilde{q}_2\| = \sqrt{1^2+(-1)^2+0^2} = \sqrt{2}$$

$$\Rightarrow q_2 = \frac{q_2}{\|\tilde{q}_2\|} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} / \sqrt{2} = \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \\ 0 \end{bmatrix}$$

$$\tilde{q}_3 = a_3 - (a_3^T q_1) q_1 - (a_3^T q_2) q_2$$

$$= \begin{bmatrix} 1 \\ 0 \\ 4 \end{bmatrix} - \left(\begin{bmatrix} 1 \\ 0 \\ 4 \end{bmatrix} \cdot \begin{bmatrix} 1/\sqrt{6} \\ 1/\sqrt{6} \\ 2/\sqrt{6} \end{bmatrix} \right) \begin{bmatrix} 1/\sqrt{6} \\ 1/\sqrt{6} \\ 2/\sqrt{6} \end{bmatrix} - \left(\begin{bmatrix} 1 \\ 0 \\ 4 \end{bmatrix} \cdot \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \\ 0 \end{bmatrix} \right) \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 0 \\ 4 \end{bmatrix} - \frac{9}{\sqrt{6}} \begin{bmatrix} 1/\sqrt{6} \\ 1/\sqrt{6} \\ 2/\sqrt{6} \end{bmatrix} - \frac{1}{\sqrt{2}} \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 0 \\ 4 \end{bmatrix} - \begin{bmatrix} 3/2 \\ 3/2 \\ 3 \end{bmatrix} - \begin{bmatrix} 1/2 \\ -1/2 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} \qquad \|\tilde{q}_3\| = \sqrt{(-1)^2+(-1)^2+1^2} = \sqrt{3}$$

$$\Rightarrow q_3 = \frac{\tilde{q}_3}{\|\tilde{q}_3\|} = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} / \sqrt{3} = \begin{bmatrix} -1/\sqrt{3} \\ -1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix}$$

Problem 5. (18 points) Ture or False (no explaination needed).

(a) If the columns of a matrix are independent, so are the rows.

False $\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$

(b) If $\mathbf{A}^T = -\mathbf{A}$ then the row space of $\mathbf{A}$ equals the column space of $\mathbf{A}$.

True

(c) The nullspace $N(\mathbf{A})$ and the row space $C(\mathbf{A}^T)$ are orthogonal complements.

True

(d) If $\mathbf{P}$ is a projection matrix, then $\mathbf{P}^2 = \mathbf{P}$.

True

(e) $\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = \begin{vmatrix} -d & -e & f \\ -a & -b & c \\ -g & -h & i \end{vmatrix}$.

False $\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = -\begin{vmatrix} d & e & f \\ a & b & c \\ g & h & i \end{vmatrix} = \begin{vmatrix} -d & e & f \\ -a & b & c \\ -g & h & i \end{vmatrix} = -\begin{vmatrix} -d & -e & f \\ -a & -b & c \\ -g & -h & i \end{vmatrix}$

(f) The determinant of $\mathbf{AB} - \mathbf{BA}$ is zero.

False $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ $B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$

$\det(AB - BA) = \det\left(\begin{bmatrix} 0 & 2 \\ -3 & 0 \end{bmatrix}\right) \neq 0$